# AI Image Generator

## Appendix

Version 2.0
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Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

# Appendix

## A. Project file tree

```
ai-image-gen/
  app.py                  web application (Gradio), theme, Admin panel
  models.py               model registry, ModelManager, LoRAs, compel
  prompt_parser.py        prompt -> settings
  sysinfo.py              system telemetry for the Admin panel
  upscaler.py             Real-ESRGAN tiled upscaling
  gpu_info.py             GPU facts and live telemetry
  db.py                   SQLite store and CLI viewer
  generate.py             command-line generator
  requirements.txt
  run-app.bat
  README.md
  Dockerfile, docker-compose.yml, .dockerignore
  generations.db          created on first generation
  models/
    RealESRGAN_x4plus.pth
    cache/                bf16 copies of converted models (auto-built, 12 GB for Z-Image)
    checkpoints/          your SDXL .safetensors files
    loras/                your SDXL LoRA files
  outputs/                generated PNG files
  docs/                   this documentation set as PDFs
  docs-src/
    build_pdf.py          source of the documentation
    images/               figures
  .venv/                  virtual environment (not for backup)
```

## B. requirements.txt

```
# Install PyTorch with CUDA first (RTX 3090 / any recent NVIDIA card):
#   pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126
# Then:
#   pip install -r requirements.txt
diffusers>=0.31
transformers>=4.45
accelerate>=1.0
safetensors
huggingface_hub
gradio>=5.0
pandas
psutil>=5.9           # Admin panel: CPU, memory, disk and network stats
spandrel>=0.4          # Real-ESRGAN upscaler
peft>=0.13             # LoRA loading
compel>=2.0            # prompt weighting (word:1.3) and long prompts for SDXL models
sentencepiece          # T5 tokenizer for FLUX
protobuf               # T5 tokenizer for FLUX
bitsandbytes>=0.45     # 4-bit FLUX on 24 GB cards
```

## C. Command cheat sheet

All commands run from the project folder in PowerShell with `.\.venv\Scripts\python.exe`.

| Task | Command |
|---|---|
| Start the app | app.py |
| List models | generate.py --list-models |
| Generate from CLI | generate.py "prompt" --model juggernaut --seed 42 --count 2 |
| Generate with a LoRA | generate.py "prompt" --model dreamshaper --lora name:0.8 |
| Upscale a file | upscaler.py outputs\x.png 3840 |
| List history / one record / statistics | db.py \| db.py 4 \| db.py stats |

| Task | Command |
|---|---|
| Parser self-test | prompt_parser.py |
| Model registry status | models.py |
| Fetch the starter LoRAs | models.py fetch-loras |
| GPU telemetry dump | gpu_info.py |
| Full system telemetry as JSON | sysinfo.py |
| Log in to Hugging Face (for FLUX) | .\.venv\Scripts\huggingface-cli login |
| Rebuild the PDFs | docs-src\build_pdf.py |
| Run in Docker | docker compose up --build |
| Check CUDA | -c "import torch; print(torch.cuda.is_available())" |

## D. Programmatic access to the running app

```
from gradio_client import Client
c = Client("http://127.0.0.1:7860")
images, report, records, sel, bar, hint, loaded = c.predict(
    "a lighthouse, 16:9, seed 42", "", "juggernaut", api_name="/generate")
table, stats = c.predict("lighthouse", 50, api_name="/refresh_history")
tiles, gpu, cpu, mem, net, power, soft = c.predict(False, api_name="/admin_refresh")
```

## E. Licences

| Component | Licence |
|---|---|
| Stable Diffusion XL base 1.0 | CreativeML Open RAIL++-M |
| Juggernaut XL v9, RealVisXL V5.0, DreamShaper XL v2 Turbo | CreativeML Open RAIL++-M |
| Playground v2.5 | Playground Community License |
| Animagine XL 3.1 | Fair AI Public License 1.0-SD |
| Z-Image Turbo, FLUX.1 schnell | Apache 2.0 |
| Real-ESRGAN weights and code | BSD 3-Clause |
| diffusers, transformers, accelerate, peft, Gradio | Apache 2.0 |
| compel, spandrel, bitsandbytes | MIT |
| psutil, PyTorch | BSD-style |
| Project source code | Owned by Frazile AI |

Each model's licence is linked from its Hugging Face page. Check the use restrictions before commercial use.